S.E.A COLLEGE OF ENGINEERING &TECHNNOLOGY

MAJOR PROJECT PRESENTATON ON.

32-BIT RISC-V CORE PROCESSOR

Project Associates

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INTRODUCTION

- RISC-V is an open-source instruction set architecture used to design modern processors.
- This project focuses on designing a 32-bit RV32I RISC-V processor using Verilog HDL.
- The processor uses a five-stage pipeline: Instruction Fetch, Instruction Decode, Execute, Memory Access, and Write Back .
- .Pipelining improves processor performance by allowing multiple instructions to be processed simultaneously.
- The designed processor is verified using simulation and testbench to ensure correct operation.

PROBLEM STATEMENT

- Design a 5-stage RV32I pipelined RISC-V processor using Verilog HDL with IF, ID, EX, MEM, and WB stages.
- Handle pipeline challenges such as data hazards, control hazards, forwarding, and stalls for correct instruction execution.
- Verify the processor core using simulation and testbenches to ensure functional correctness and reliable performance.

OBJECTIVES

- To design and implement a 32-bit RV32I RISC-V processor core using Verilog HDL with a five-stage pipelined architecture.

- To improve processor performance by executing multiple instructions simultaneously through Instruction Fetch, Decode, Execute, Memory, and Write Back stages.

- To verify the functionality and correctness of the designed processor using testbench-based simulation and waveform analysis

LITERATURE SURVEY

Table 1:literature survey

Author & Year	Title of Paper	Methodology Used	Limitations
1. E. Vinothkumar et al., 2026	Design and Implementation of 32 Bit Single Cycle Core Based on RISC-V ISA	Designed a 32-bit RV32I single-cycle processor using Verilog HDL and verified its operation through FPGA implementation.	Single-cycle architecture provides lower performance compared with pipelined architectures.
2. K. Kırali et al., 2025	Implementation of FPGA Based 32-bit RISC-V Processor	Designed and implemented a compact 32-bit RISC-V processor on an FPGA, focusing on efficient hardware utilization and power consumption.	FPGA resources restrict optimization compared with a dedicated ASIC implementation.
3. Hussain et al., 2024	FPGA Implementation of Five Stage Pipelined RISC-V Processor	Developed a 32-bit RISC-V processor using a five-stage pipeline and implemented it on FPGA for functional evaluation.	Pipeline hazards require additional forwarding, detection and control logic.
4. A. Tiwari et al., 2023	Design and Implementation of a Pipelined RISC-V Processor	Designed a 32-bit RV32I pipelined processor and evaluated its performance through hardware implementation and simulation.	Increased pipeline stages improve performance but also increase hardware and control complexity.

1. PROBLEM DEFINITION

Define the objectives and requirements for designing a 32-bit RISC-V core processor (RV32I).

2. ARCHITECTURE STUDY

Study RISC-V RV32I ISA specification, instruction set, and processor architecture.

3. DESIGN USING VERILOG HDL

Design the processor modules in Verilog HDL based on the RISC-V architecture.

4. 5-STAGE PIPELINE DESI

Implement 5-stage pipeli
for high performance.

IF → ID → EX → MEM →

5. MODULE INTEGRATION

- Register File
- ALU
- Control Unit
- Memory Interface
- Hazard Detection & Forwarding
- Pipeline Registers

6. SIMULATION

Simulate the design using testbench in ModelSim / Vivado Simulator.

7. VERIFICATION

Verify functional correctness of the processor using test programs and instruction set validation.

8. RESULT ANALYSIS

Analyze performance, area, and power. Validate the design and document the results.

SOFTWARE REQURIMENTS

Software Requirement

- Verilog HDL
- Verilator / ModelSim
- Vivado / Quartus
- RISC-V GCC Toolchain
- Spike Simulator

HARDWARE REQUIREMENTS

- FPGA board (Xilinx /Intel/Lattice)
- 4 GB RAM minimum
- Dual-core processor or higher

EXPECTED OUTCOMES

- Design and implement a functional 32-bit RISC-V processor core using Verilog HDL.
- Verify the processor through simulation and testing to ensure correct execution of RISC-V instructions.
- Achieve efficient processor operation with improved performance, reliable data processing, and proper pipeline execution.

ADVANTAGES

- High performance
- Open-source architecture
- Simple and modular design
- Low power consumption
- Easy to customize

DISADVANTAGES

- Pipeline design is complex
- Verification is time-consuming
- Limited software ecosystem
- Hazard handling is difficult

APPLICATION

- Embedded systems
- IoT devices
- Industrial automation
- Consumer electronics
- Educational and research platforms

REFERENCES

1) Patterson, D. A., & Hennessy, J. L. (2021), Computer Organization and Design: The Hardware/Software Interface, RISC-V Edition.

2) Harris, D. M., & Harris, S. L. (2021), Digital Design and Computer Architecture: RISC-V Edition, covering processor and pipeline design concepts.

3) Waterman, A., & Asanović, K. (2019), The RISC-V Instruction Set Manual, providing detailed information about RISC-V instruction execution.

4) Palnitkar, S. (2003), Verilog HDL: A Guide to Digital Design and Synthesis, explaining Verilog-based digital system design.

5) Sutherland, S., Davidmann, S., & Flake, P. (2019), SystemVerilog for Design, covering RTL design and hardware verification techniques.

6) Asanović, K., & Patterson, D. A. (2014), Instruction Sets Should Be Free: The Case for RISC-V, explaining the motivation and benefits of the RISC-V architecture.

7) Waterman, A. (2016), Design of the RISC-V Instruction Set Architecture, discussing the design principles and implementation of RISC-V.

8) Codasip (2022), RISC-V Processor Design and Architecture, covering customizable RISC-V processor design and implementation.

9) David A. Patterson (2021), Computer Organization and Design: RISC-V Edition

THANK YOU